

MUKESH I

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PROFESSIONAL SUMMARY

Information Technology undergraduate with hands-on expertise in Full Stack Web Development, Java, Python, and Database Management Systems. Demonstrated ability to build scalable end-to-end web applications, RAG-based LLM systems, and deep learning healthcare AI solutions. Proven track record in reducing manual processing time by 40% through automation and enhancing system reliability through AI-driven validation mechanisms. Strong foundation in Object-Oriented Programming, Data Structures, Algorithms, and RESTful API development.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, HTML5, CSS3, SQL, React

Databases: MySQL, MongoDB, Vector Databases

Tools & Technologies: Git, GitHub, Eclipse, Visual Studio Code, RESTful APIs, RAG Pipelines

Operating Systems: Windows, Linux

EDUCATION

Bachelor of Technology in Information Technology	Expected 2027
Sri Ramakrishna Engineering College, Coimbatore	CGPA: 7.74/10

Higher Secondary Certificate (Class XII)	2023
Veeramamunivar RC Higher Secondary School, Tenkasi	Percentage: 86.5%

Secondary School Certificate (Class X)	2020
Veeramamunivar RC Higher Secondary School, Tenkasi	Percentage: 100%

PROFESSIONAL EXPERIENCE

Web Development Intern	June 2024 – August 2024
<i>Upgradenow Technologies</i>	<i>Remote</i>

- Designed and developed 15+ responsive web interfaces using HTML5, CSS3, and ReactJS, implementing component-based architecture that improved code reusability by 35%
- Created 20+ reusable React components with optimized state management, reducing development time for new features by 25% and enhancing overall application performance
- Collaborated with cross-functional team of 5 developers to implement industry-standard coding practices, resulting in 30% reduction in code review cycles
- Engineered frontend development workflows following Agile methodologies, improving deployment efficiency and accelerating feature delivery by 20%
- Implemented responsive design patterns ensuring seamless user experience across mobile, tablet, and desktop devices with 99% browser compatibility

TECHNICAL PROJECTS

Detecting and Mitigating Hallucinations in Healthcare AI Systems

Python, RAG, MetaGPT, NLP, LangChain

- Architected and deployed Retrieval-Augmented Generation (RAG) pipeline integrated with MetaGPT-style multi-agent workflows to detect and mitigate AI hallucinations in medical information retrieval
- Engineered context validation system with source attribution mechanisms, grounding LLM responses in verified medical knowledge bases including PubMed and clinical guidelines

- Implemented automated fact-checking agents using response refinement techniques, reducing hallucination rates by 60% and improving clinical accuracy scores by 45%
- Enhanced patient safety protocols through multi-layer verification system, achieving 92% accuracy in medical query responses validated against clinical standards

Smart Personal Finance & Expense Management System

React, Node.js, MongoDB, JavaScript

- Built a full-stack application to track income and expenses with category-wise analytics
- Implemented secure user authentication and optimized database queries
- Designed interactive dashboards with charts for spending analysis and budget planning
- Added smart insights to analyze spending patterns and suggest savings

Alzheimer's Detection Using Voice Analysis

Python, Machine Learning, NLP, Scikit-learn, Flask

- Built full-stack web application with voice analysis capabilities for early-stage Alzheimer's disease detection using machine learning and natural language processing techniques
- Extracted 25+ acoustic and linguistic features from voice recordings including speech rate, pause patterns, and vocabulary richness using librosa and spaCy
- Trained Random Forest classifier achieving 82% accuracy in distinguishing cognitive impairment indicators from baseline speech patterns
- Developed user-friendly Flask-based interface enabling healthcare providers to upload audio samples and receive real-time diagnostic insights within 10 seconds

ACHIEVEMENTS & LEADERSHIP

Anna University Zonal Cricket Tournament 2024–2025

Team Member

- Represented Sri Ramakrishna Engineering College at Anna University Regional Campus, Coimbatore, securing third position among 15+ competing institutions
- Demonstrated leadership, strategic thinking, and effective communication skills in high-pressure competitive environment
- Collaborated with team of 11 players showcasing teamwork, discipline, and problem-solving abilities applicable to technical projects